COMSATS University Islamabad Lahore Campus

Mid-Term Examination – SPRING 2024

Course Title:	Database Systems				Course Code:	CSC270	Credit	3+1
Course	Abdul Qayyum				Programme	BSSE		
Semester:	4th	Batch:	FA22	Section:	A, B	Date:	24-04-24	
Time Allowed:	90 Minutes				Maximum Marks:		25	
Student's					Reg. No.			

Important Instructions / Guidelines:

Question 1: [CLO-1] [5]

Explain the following terms with example:

a) *Domain constraints*
b) *Entity Integrity*
c) *Referential Integrity*
d) *Base Relation*
e) *Virtual Relation*

Question 2: [CLO-2] [5+5=10]

a) Why we use Restricted Cartesian Product instead of Cartesian Product. Explain with an example.
b) What is the difference between Logical Database Design and Physical Database Design?

Question 3: **Draw an ERD of following textual descriptions** [CLO-3] [10]

Comsats University is developing a database to streamline its admission system. The admission at Comsats University is held twice a year, in the Summer and Fall semesters. When the admission for a specific semester opens, students are required to submit their applications within the due date. Students have the flexibility to indicate up to three preferred degree programs when submitting their applications for admission. Essential to the admission process is the National Testing Service (NTS) test for each department. Students are afforded the opportunity to undertake the NTS test multiple times, allowing them to enhance their scores.

Merit for admission is precisely calculated, taking into account various parameters including the applicant's marks %age in their Secondary School Certificate (SSC) or equivalent (Matric), Higher Secondary School Certificate (HSSC) or equivalent (FSC), and the results of the NTS test. The merit list for each degree program is subsequently compiled based on available seats in each degree program, with preference given to candidates with higher merit rankings. Admission offers are extended to students in accordance with the merit calculations.

Students receiving admission offers are presented with a preference-wise list of degree programs, allowing them to select their desired program based on their ranking and eligibility. Upon acceptance of the admission offer, students are issued fee vouchers and are required to submit their dues to designated banks such as Habib Bank Limited (HBL) or Allied Bank Limited (ABL). If seats remain unfilled following the initial merit list, subsequent merit lists are generated and displayed to offer admission opportunities to additional candidates. The iterative process continues until all available seats are filled or until the specified admission deadline is reached.

CUI
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